Pizza Tomato Sauce Masterclass: An In-Depth, Technical Guide

While the dough provides the structure, the tomato sauce is often what defines the personality of a pizza. With a unique balance of sweetness, acidity, and umami, a good sauce brings the crust and cheese into harmony. This comprehensive guide covers everything you need to know about pizza tomato sauce – from the science of canned tomatoes and the different products available, to how the sauce affects both the bake and the final pizza. You will also find practical tips and techniques for controlling flavor, texture, and consistency to help you create your own perfect sauce

Pizza Tomato Sauce: An Introduction

Tomato sauce is one of the most important ingredients in pizza – possibly the one that affects its flavor the most. It brings everything together, helping the dough, cheese, and toppings work in harmony.

A classic pizza is made up of three main parts: dough, cheese, and tomato sauce.

The **dough** is the base of the pizza, quite literally. It holds the other ingredients and gives the pizza its **texture**. It also adds a lot of flavor, especially after it ferments and develops flavors over time.

Cheese – usually **mozzarella** – also adds [some] flavor, but its role is more limited. Mozzarella has a **very mild taste**, so it mostly adds **texture**, especially in how it melts and stretches when baked.

Tomato sauce, made from tomatoes of course, brings a bold and noticeable tomato flavor. That's because tomatoes are high in **glutamic acid**, which gives food its "umami" – a deep, savory taste (and is also what MSG – monosodium glutamate – is made from).

Interestingly, the jelly-like part around the tomato seeds has 4 to 6 times more glutamic acid than the rest of the tomato, or in simpler terms – packed with flavor.

Because tomatoes have a strong and dominant flavor, the sauce often has the biggest impact on both the flavor and texture of the pizza:

- **Effect on Flavor:** The sauce can either hide or highlight the flavors of the dough and cheese.

- The **type of sauce** used plays a major role in shaping the overall taste of the pizza.
- **Effect on Texture:** The amount and consistency of the sauce directly affect the pizza's eating characteristics – whether it's wet or dry, heavy or light, and even how crispy the crust gets.
- The texture of the sauce – whether smooth or chunky – also plays a part in the final texture of the pizza.
- **Effect on Cheese:** The consistency of the sauce can change how the cheese bakes.
- **Layering of the Sauce:** Whether the sauce is placed above or below the cheese changes the look of the pizza, how the cheese melts, and how crispy the base becomes, and the overall eating experience of the pizza.

While the dough adds a lot to the overall flavor of the pizza, for most people – especially those who aren't focused on the finer details – the **sauce** is often what "defines" the pizza.

When tasting a pizza as a whole, especially from a **flavor perspective alone**, the sauce often stands out the most, and can easily overshadow the dough, largely shaping the flavor profile of the entire pizza.

A good pizza has a tomato sauce that **complements the other ingredients and works together with them**, matching the overall flavor the baker is aiming for (more on that in the next section). The sauce that one person enjoys might not appeal to someone else – it's all about personal taste.

Tomato sauce also plays a key role in **setting pizzas apart**. A sauce made from different tomato products, with different seasonings, textures, consistencies, and amounts, will result in a completely different pizza. In simple terms: **the sauce helps give the pizza its unique "identity."**

Defining the Sauce Flavor Profile

There are many ways to make pizza sauce, and the choice of method, ingredients, and tomato products all shape its flavor. This, in turn, affects the overall taste and eating experience of the pizza.

So, the first question to ask when making pizza sauce is: **What kind of flavor profile are you looking for?**

Here are some common types of pizza sauce, each with its own distinct flavor profile:

- **Sweet:** A sauce where sweetness stands out, either from naturally sweet tomatoes or added sweeteners like honey or sugar.
- **Acidic/Tangy:** A sauce with a noticeable tanginess, highlighting the natural acidity of tomatoes.
- **Strong and Intense:** A sauce with a bold tomato flavor (relying on naturally flavorful tomatoes rather than heavy seasoning).
- **Mild:** A sauce with a more delicate tomato taste, either made with mild tasting tomatoes or thinned out sauce for a lighter flavor.
- **Fresh and Bright ('Plain' Sauce):** A sauce that keeps the bright, raw taste of fresh tomatoes, with minimal seasoning.
- **Concentrated/Cooked:** A sauce with a deeper, richer tomato flavor and less freshness, achieved by cooking it down or using tomato paste or other concentrated tomato products.
- **Seasoned:** A sauce where the added herbs and spices play a major role in the final flavor.

These categories can, of course, overlap. For example, a fresh sauce can be sweet or acidic, depending on the tomatoes used; a concentrated sauce can also be seasoned; and a fresh sauce can still have some added seasoning.

In general, pizza sauces fall into two main categories (which may also overlap):

- **Seasoned vs. plain (unseasoned) sauce**
- **Concentrated/cooked vs. fresh sauce**

Seasoned vs. Plain (Unseasoned) Sauce

This is the main distinction when it comes to sauce flavor profile: a sauce that highlights **fresh, bright** tomato flavor, versus one where **seasoning** takes the lead. These two styles generally don't overlap.

In a **plain sauce**, the focus is entirely on the tomatoes. It is **minimally seasoned**, usually just with salt, and sometimes a little olive oil or fresh basil, to enhance the tomato's natural flavor without overpowering it.

The result is a sauce that tastes fresh and vibrant, making the quality of the tomatoes crucial. For this reason, **a fresh sauce requires high-quality tomato product**, with a naturally rich flavor.

A classic example of a plain sauce can be found in Neapolitan pizza and many famous New York-style pizzerias, where the sauce is made from high-quality canned tomatoes and has a simple, fresh yet excellent taste.
In contrast, a **seasoned sauce** goes beyond basic salt seasoning. Additional herbs and spices create a more complex flavor, often masking the tomato's natural brightness and fresh flavor (for better or worse, depending on personal preference). Some even argue that heavily seasoned sauces resemble pasta sauce more than pizza sauce.

Seasoning can also help mask or "smooth out" the flavors of lower-quality tomato products, that may not taste great on their own. For example, Pizza Hut's sauce, as well as many chain pizzerias serving American-style pizza, is typically heavily seasoned and made with tomato paste.

Later in this article, you'll find a list of common seasonings that can be added to pizza sauce.

Fresh (Uncooked) vs. Concentrated (Cooked) Sauce

A **concentrated sauce** has one or both of the following characteristics:

1. It has been **cooked** to concentrate it and intensify flavors.
2. It is made with or **mixed with concentrated tomato products**, such as tomato paste.

Cooking the sauce or using concentrated tomato products reduces the fresh, bright notes of the tomatoes, creating a **deeper, more complex sauce, with a richer, less "fresh" flavor profile**.

Therefore, If you aim for a sauce that highlights the **freshness of tomatoes**, avoid cooking it or using concentrated tomato products.

On the other hand, if you prefer a **deeper, more developed tomato flavor**, consider using tomato paste as the base of your sauce, mixing it with other tomato products (covered in the next section), or cooking the sauce until it thickens and reaches your desired concentration.

Selecting the Best Tomato Products for Your Sauce

Before going into the details of making pizza tomato sauce, it's important to understand the different tomato products available and how they differ.

There are three main types of tomato products used for making tomato sauce (not just for pizza):

1. **Canned Tomatoes (Crushed/Diced/Whole):** These are fresh tomatoes preserved through canning immediately after harvest. This includes any tomato product sold in a can or jar. **Canned tomatoes are the most common choice for pizza sauce**, widely used by home bakers and top pizzerias alike.
2. **Concentrated Tomato Products (Tomato Paste/Sauce):** These tomatoes are cooked down into a thick, concentrated product with an intense flavor. Tomato paste can be diluted with water to form a base for sauce, or added to enhance an existing sauce. Many chain pizzerias, such as Pizza Hut, Domino's, and Papa John's, use tomato paste-based sauces, which tend to have a strong, bold flavor.
3. **Fresh Tomatoes:** Fresh, ripe tomatoes processed into a sauce. This is the least common option for pizza sauce, and is often not recommended (more on that later).

Technically, tomato paste falls under the category of "canned tomatoes", since it also undergoes a preservation process.

Why Canned Tomatoes are the Industry Gold Standard

Canned tomatoes (or **tinned tomatoes**, as they're called in the UK), are the **most common and recommended tomato product for making pizza sauce** (and tomato sauce in general). In the following sections, we'll cover the basics of canned tomatoes and review the different types available.

Understanding the Canned Tomato Industry

This section applies to all types of canned tomatoes, including tomato paste and passata/tomato puree.

Surprisingly, **canned tomatoes are often the freshest form of tomatoes available for making tomato sauce**. This is because they are harvested at **peak ripeness**, and immediately preserved through canning. This process ensures that canned tomatoes, in all their varieties, are **much fresher and flavor-packed** than most tomatoes found in supermarkets or produce stores (greengrocers).

Most tomatoes available in stores are **picked while still green and not fully ripe**. After being harvested, they undergo an "artificial" ripening process using ethylene gas to turn red. The reason for picking them green is purely logistical – ripe

tomatoes are delicate and would easily crush or spoil during transport, while green, firm tomatoes are much more resilient.

Although these tomatoes may appear red, they often **lack the rich flavor that develops during natural vine ripening**. In contrast, **canned tomatoes are vine-ripened and preserved at their peak of flavor**.

For this reason, **canned tomatoes are the best choice for making tomato sauce**, as they offer maximum flavor.

The only exception to the above is if you have access to fresh, vine-ripened tomatoes, such as those bought directly from a farmer, or home-grown.

How Canned Tomatoes are Produced

The production process for all canned tomato products follows a similar series of steps:

1. **Picking at Peak Ripeness**: Tomatoes are harvested only when **fully ripe**. Plum tomatoes (Roma tomatoes) are commonly used, due to their higher flesh-to-liquid ratio, making them ideal for canning.
2. **Transport and Immediate Preservation**: The tomatoes are quickly transported to the factory and processed within hours to maintain freshness.
3. **Cleaning**: The tomatoes are washed thoroughly to remove any dirt, insects, or debris.
4. **Scalding**: The tomatoes are briefly boiled to loosen the skins for easy removal.
5. **Peeling**: The skins are removed (though some products may retain them intentionally).
6. **Processing into the Final Product**: The tomatoes are processed into different forms, such as crushed, diced, whole, puree, or paste.
7. **Heating and Sterilization**: The tomatoes are heated at 80-100°C (176-212°F) to kill bacteria. The heating duration varies based on the type of tomato product.
8. **Packaging**: The tomatoes are vacuum-sealed in cans (or jars) to prevent bacterial growth.
9. **Final Heating**: The sealed cans undergo a final sterilization process through hot water baths or pressurized heating.

10. **Distribution**: The products are cooled, packed, and sent to stores.

Note that canned tomatoes undergo three stages of heating: first during scalding, then before being processed into their final form, and finally after packaging. The duration and intensity of each stage vary depending on the production method and the specific type of tomato product.

It's important to note that **canned tomatoes contain no preservatives**. Their long shelf life and freshness are preserved through complete sterilization and vacuum sealing, eliminating the need for artificial additives.

Technical Differences Between Canned Products

Canned tomato products vary significantly based on several factors, whether from the same manufacturer or different brands:

- **Tomato quality** (variety, type, overall freshness, etc.).
- **Type of product** (whole, crushed, diced, paste, etc.).
- **Additional ingredients** beyond tomatoes.
- **Production process** and methods used.
- **Concentration level** of the tomatoes, measured by degree Brix (°Bx).

Source Quality and Tomato Variety

The most significant factor affecting the taste and quality of canned tomatoes is the **quality of the tomatoes themselves**. Simply put, the more flavorful the tomatoes a manufacturer uses, the better the final product.

Since tomatoes are a natural, seasonal product, harvest conditions also influence quality. Even for the same brand and product, variations can occur from season to season based on the year's tomato crop.

Different manufacturers source tomatoes of varying quality, which is the primary reason why canned tomato products differ in flavor, even within the same brand. High-quality canned tomatoes have a balance between acidity and sweetness, while offering a strong aroma and fresh tomato flavor.

Of course, taste is subjective, and it's best to experiment with different products to find the one you prefer.

For example, canned San Marzano tomatoes, considered premium tomato product, are generally more expensive than those made from 'regular' tomato varieties. However, price does not always equal quality, and preference ultimately depends on

personal taste.

Packaging Methods: Juice vs. Purée Packs

Another key difference among canned tomato products is how the tomatoes are packed. This can generally be divided into three categories:

- **100% crushed tomatoes**: These contain only tomatoes, usually crushed or diced, without any added juice, puree, or concentrate.
- **Tomatoes in paste or concentrate**: These are usually crushed tomatoes packed in a concentrated tomato base, such as paste or puree.
- **Tomatoes in juice or puree**: Typically whole or diced tomatoes packed in tomato juice or a thinner puree.

The label will typically specify how the tomatoes are packed (e.g., "crushed tomatoes in puree" or "whole tomatoes in tomato juice").

To know exactly what you're getting, check the ingredient list – it will indicate the percentage of tomatoes, and any additional components like juice, puree, or paste.

As we'll discuss later, how the tomatoes are packed has a noticeable effect on both the **texture** and **flavor** of the final sauce – especially if you're using the product without diluting or thickening it. For example:

- 100% crushed tomatoes have a different flavor profile compared to crushed tomatoes with added tomato paste.
- Whole or cubed tomatoes in juice tend to preserve a fresher, brighter tomato flavor. When combined with their juice, they produce a different taste compared to crushed products (even of the same company).

Additives: Citric Acid and Calcium Chloride

Some canned tomato products include ingredients beyond just tomatoes:

- **Puree / Juice / Paste** – as detailed in the previous section.
- **Salt** – commonly added for flavor and preservation.
- **Sugar** – added to enhance sweetness.
- **Calcium Chloride (E509)** – found only in tomatoes packed in juice. This is a firming agent that helps the tomatoes retain their shape and prevents them from becoming mushy while in the juice.
- **Citric Acid** – a natural preservative that occurs in citrus fruits and tomatoes. It helps preserve the tomatoes' bright red color and boosts acidity, especially if the tomatoes are naturally low in it.

- **Spices/Seasoning** – typically found in ready-made pizza sauces.

Generally speaking, higher-quality canned tomato products contain **only tomatoes and salt**. The addition of sugar or citric acid may suggest that the tomatoes lack sufficient natural sweetness or acidity on their own.

That said, taste is what matters most; if you enjoy a particular product – use it.

The Impact of the Canning Process

Different manufacturers use different canning and preservation methods, which can significantly affect the final flavor of the tomato product – for better or worse.

Understanding Concentration Levels (Degrees Brix)

Degrees Brix (°Bx) measures the **concentration of dissolved solids in a tomato product** – primarily **sugars**, but also acids, minerals, and other solids.

For example, a product labeled as 28°Bx contains 28% tomato solids and 72% water.

In simple terms, the **higher** the °Bx value, the **more concentrated** the product. A higher concentration means less water and more tomato solids, resulting in a **deeper, more intense flavor**. This also tends to reduce the "fresh" tomato taste. Whether this is desirable depends on your intended use and flavor preference.

Here's a general Brix range for various tomato products:

- Fresh tomatoes: 4–6°Bx
- Tomato juice: 5–7°Bx
- Tomato puree, passata, whole/diced tomatoes in juice: 7–12°Bx
- Crushed tomatoes: 10–15°Bx
- Tomato paste/sauce (concentrated): 18–38°Bx
- Sun-dried tomatoes: 20–30°Bx

Understanding Specific Canned Tomatoes Products

In the following sections, we'll explore the different types of canned tomato products available.

Whole Peeled, Crushed, and Diced Tomatoes

These are the **most commonly used tomato products for pizza sauce**. They are minimally processed and come [packed] in several forms:

- Crushed/chopped/ground tomatoes (also called 'polpa di pomodoro') – either 100% crushed tomatoes, or with added tomato paste or puree.
- Diced tomatoes in tomato juice or puree.
- Whole tomatoes in tomato juice or puree.

Crushed tomatoes are in most cases the most convenient option, as they can typically be used straight from the can with little to no additional processing, just seasoning as needed.

Whole and diced tomatoes in juice, on the other hand, require extra preparation – they must be blended, crushed, or chopped manually to achieve sauce consistency (more on that later).

Tomato Paste and Concentrates

Concentrated tomato products are made by reducing the liquid content of tomatoes, leaving behind a **rich, deep, and intensely flavored paste**. Due to the higher concentration of natural sugars, these products often have a notably sweet(er) taste.

The production process of concentrated tomato products involves cooking tomatoes into a thick paste, filtering out excess liquids, and further reducing moisture through evaporation to intensify the tomato flavor.

Tomato paste can serve as a base for pizza sauce, creating a bold and robust flavor. It can also be added to sauces made with non-concentrated tomato products to enhance depth and richness, and/or to thicken the sauce's consistency.

Tomato Purée and Passata

Tomato puree, or **passata** in Italian, is a **smooth, seedless, and skin-free tomato product**, with a paste-like consistency.

During tomato puree production, tomatoes are crushed and filtered to remove seeds and solid pieces, resulting in a **smooth, velvety texture**.

Tomato puree/passata products are typically more concentrated than crushed tomatoes, but less concentrated than tomato paste or other concentrated tomato products. In general, products labeled as "passata" tend to be **less** concentrated than those labeled "tomato puree."

Tomato puree can be used as a base for pizza sauce, or combined with other tomato products to adjust texture and flavor.

Note that not all tomato puree products are born equal:

– Some are made by **directly processing fresh tomatoes** – crushed and filtered to a smooth consistency – without first being concentrated into paste and then diluted with water. For example, Stanislaus' tomato puree products are made directly from fresh, vine-ripe tomatoes.
– Others are made from **reconstituted tomato paste** (i.e., tomato paste diluted with water), which results in a flavor profile closer to concentrated tomato products than to fresh tomatoes.
– Some puree products are more concentrated than others, as there's no strict regulation defining what qualifies as "tomato puree."

It's worth checking the ingredient list or labeling to understand which type you're getting. If a tomato puree product's ingredient list says "Water, Tomato Paste," then it's made from reconstituted tomato paste.

A Note on Store-Bought "Pizza Sauce" Products

Most supermarkets carry a variety of ready/pre-made pizza sauces, each with a different composition. These sauces are typically made from crushed tomatoes, tomato paste, or a combination of both, along with various seasonings.

The ingredient list will specify whether the sauce is based on tomato puree, concentrate, or crushed tomatoes, as well as any added seasoning, giving you an idea of what flavor profile to expect.

While ready-made pizza sauce is convenient, it is often more expensive than homemade sauce, and limits control over flavor and seasoning.

Using Fresh Tomatoes to Make Pizza Sauce

Is Making Sauce from Fresh Tomatoes Worth the Effort?

Making pizza sauce from fresh tomatoes is possible, **but it's only worthwhile if you have access to vine-ripened tomatoes**, rather than the standard supermarket or produce stores varieties.

As explained earlier in this article, canned tomatoes are picked at peak ripeness, **while supermarket and produce stores tomatoes are harvested green**, resulting in **significantly less flavor**. Simply put, even half-decent canned tomatoes, being vine-ripened, will yield a **far more flavorful sauce** than typical supermarket tomatoes.

In most cases, using non-vine-ripened fresh tomatoes is, to be blunt – **a waste of time**. The resulting sauce will be inferior to one made with decent-quality canned tomatoes.

However, if you have access to vine-ripened tomatoes – whether from a local farmer or your own garden – and want to use them to make sauce, opt for **plum tomatoes** (also called Roma tomatoes), as they are fleshier and contain less liquid.

Additionally, choose the **ripest tomatoes possible**, as they tend to develop more flavor and are typically sweeter and easier to process into sauce.

A Step-by-Step Guide to Processing Fresh Tomatoes Into Pizza Sauce

Here are the steps for processing fresh tomatoes to make pizza sauce:

1. Calculating Required Raw Weights

Before making the sauce, it's important to determine how many tomatoes you'll need.

As a general rule of thumb, when using **plum** tomatoes, they will yield anywhere between 60-90% **raw** tomato sauce (after straining out excess juice/water), depending on the tomato variety, ripeness, and other factors. This means that 100 grams of tomatoes will yield anywhere between 60-90 grams of raw tomato sauce.

A standard plum tomato weights between 70-150 grams, depending on its specific variety and size.

Additionally, most fresh tomato sauces require **some cooking and thickening**, which further reduces the final yield as water evaporates.

Bottom line: Expect to use **1.5-3 times** the weight of fresh tomatoes for the final sauce quantity you're aiming for.

2. Scalding and Peeling

The first thing to do when using fresh tomatoes is to briefly **scald** them in hot water to soften the skin for easy removal.

Follow these steps to scald the tomatoes:

1. Boil water in a sufficiently large pot.
2. Prepare a separate bowl filled with ice water or very cold water. The tomatoes will be transferred to this container immediately after scalding to stop the cooking process.
3. Slash the bottom of each tomato in an X shape (not on the stem side). This incision will facilitate peeling the skin after scalding. The cut doesn't need to be deep – just enough to break the skin (see image below).
4. Place the tomatoes in the hot water and leave them for no more than one minute.
5. Remove the tomatoes from the pot and transfer them directly to the bowl with cold water. Let them sit in the cold water for about one minute or until they are cool enough to handle.
6. Peel the skin from the bottom of the tomato (where the X cut is) – it should come off easily.

3. Initial Processing and Seed Removal

After scalding, there are several ways to process the tomatoes into sauce:

1. Using a blender
2. Using a food mill
3. Finely chopping with a knife

Using a Blender

Generally, using a blender is the **least recommended** method for processing fresh tomatoes into sauce, as it causes changes in texture and flavor (more on this later).

If you must use a blender or stick blender, do so in **short pulses and as minimally as possible**.

Processing in a Food Mill

A food mill is the most effective way to process tomatoes, producing a **smooth and uniform sauce while removing seeds and skins**. I *highly* recommend using a food mill for making tomato sauce, even if you're working with canned tomatoes (more on this later).

Chopping with a Knife

Chopping fresh tomatoes with a knife is the next best option after using a food mill. Simply chop the tomatoes finely until you achieve a relatively smooth and uniform texture.

4. Final Reduction and Consistency Adjustment

Now that we have raw tomato sauce, it can be used in two ways:

- As is, with no further processing.
- Cooked to reduce the liquid and concentrate the flavor.

In most cases, depending on the type of tomatoes used, a sauce made from fresh tomatoes **will need to be cooked**; otherwise, it will be too watery and have a too-mild flavor due to its high liquid content.

To cook the sauce, simmer it in a saucepan over low heat for 5 to 30 minutes, stirring and tasting occasionally until you reach your desired flavor and consistency.

Practical Tips for Mastering Your Pizza Sauce

In the following sections, you will find practical tips for making [tomato] pizza sauce, specifically covering:

- Controlling the flavor profile of the sauce
- Controlling the texture and consistency of the sauce
- The layering of the sauce (above or below the cheese) and its effect on the pizza
- Whether to cook the sauce or not
- How to determine the amount of sauce to use
- Tips for reducing heartburn from tomato sauce
- Preparing the sauce in advance and allowing it to rest to develop flavors
- The impact of raw garlic and onion on the sauce
- How to store and freeze leftover sauce

Controlling the Flavor Profile

The flavor profile of the sauce consists of three main factors:

- **The base of the sauce**: whether it is a concentrated tomato product, a ‘fresh’ tomato product (crushed tomatoes, whole tomatoes, etc.), or a combination of both.
- **The specific tomato product used**: different brands and types of tomatoes can have varying levels of sweetness, acidity, and depth of flavor.

- **The seasonings added to the sauce**: including salt, herbs, spices, garlic, onion, and other ingredients that can enhance or alter the final flavor.

As you read the following sections, keep in mind the types of sauces described at the beginning of the article – this will help provide context.

How the Base Tomato Product Shapes the Flavor

The first step in making tomato sauce for pizza is deciding whether to use a **concentrated tomato product** or a **"fresh" tomato product** as the base.

"Fresh" tomato products include **canned** tomatoes that have undergone minimal processing (such as crushed, diced, whole, or passata/puree), *or* fresh, ripe tomatoes.

Using a **concentrated** tomato product as the base will create a sauce with a **bold, rich flavor** and a natural sweetness due to the higher concentration of tomato sugars.

In contrast, a sauce made from "**fresh**" tomato products will have a **lighter, brighter** flavor, closer to that of fresh tomatoes.

If using a concentrated tomato product, **dilution is often necessary** to prevent an overpowering flavor. The standard dilution ratio is **1.5-3 parts water per 1 part sauce** (e.g., for 100g tomato paste, add 150-300g water).

Tomato juice can also be used for dilution Instead of water.

When using a **fresh tomato product**, consider whether it contains added tomato paste or juice:

- Tomatoes (whole or diced) in tomato juice: The juice is **highly flavorful** and can be used to dilute the sauce after processing, or saved for another use.
- Tomatoes packed in puree or concentrate: The amount of puree or concentrate affects the overall flavor intensity straight from the can.

To adjust the flavor, you can **combine different tomato products**, for example: adding a small amount of tomato paste to a fresh tomato product for more depth, or vice versa.

The Specific Brand Used

The specific tomato product or manufacturer you choose has the most significant impact on the flavor of the sauce. Different manufacturers use different tomato varieties and processing methods, meaning that even products from the same brand can yield noticeably different flavors.

So, what is the “best” tomato product for pizza sauce? The simple answer is: the one that tastes best **to you**. Experiment with different tomato products to find your preferred balance of sweetness, acidity, and richness.

The Art of Seasoning and Balancing Acidity

Seasoning enhances the sauce beyond the natural flavor of the tomatoes.

One seasoning that is always essential is **salt**, which not only intensifies the tomato flavor, but also balances the overall taste of the sauce. Without salt, even a high-quality sauce can taste bland or flat.

As a general guideline, aim to add at least **1% salt** based on the total weight of the sauce. For example, for 28oz (800g) of sauce, this would be 8 grams of salt (800 × 0.01 = 8), roughly two teaspoons of salt. Of course, you can adjust the salt level to suit your taste.

Keep in mind that some canned tomato products already contain salt (check the ingredients list), so you may need to reduce the amount of additional salt accordingly.

Beyond salt, a variety of other seasonings can be added to customize the sauce. In the next section you will fine different seasoning options and their effects on the final flavor.

Sauce Seasoning Options

Below is a list of seasonings that can be added to pizza sauce.

It’s best to start with a small amount of seasoning and adjust gradually – you can always add more, but you can’t remove what’s already in the sauce.

Plain Sauce with Minimal Seasoning (Base Sauce)

- **Salt** – Start with 1% of the sauce weight (e.g., 8g per 800g of sauce), adding gradually and tasting as you go.
- **Olive oil** – A small amount enhances flavor and texture.

- **Sugar (optional)** – Depending on the natural sweetness of the tomatoes and your preference.
- **Fresh basil leaves (chopped/torn)** *or* **dried oregano** – Basil provides a fresh, bright flavor, while dried oregano adds a deeper, earthier flavors.

Additional Seasonings

Aromatics:

- Fresh garlic (grated/chopped) – Start with one clove per can of tomatoes and adjust as needed.
- Garlic powder – Milder than fresh garlic.
- Fresh onion or caramelized onion – Adds depth to the sauce.
- Onion powder – A more subtle onion flavor.

Herbs & Spices:

- Dried marjoram
- Dried basil
- Dried thyme (use sparingly as it can be overpowering)
- Ground fennel seeds – Adds a mild anise/licorice note.
- Paprika – Can add a mild sweetness or smokiness.
- Black pepper (use sparingly as it can overpower the tomato flavor)
- Ground chili / chili flakes – Adds spiciness.

You can also use fresh versions of dried herbs. When substituting, use more fresh herbs than dried, as **dried** herbs have a stronger, more concentrated flavor.

Umami Enhancers:

- MSG – Enhances umami.
- Fish sauce – A natural source of umami.
- Worcestershire sauce – Adds a complex umami depth.

Acidity Enhancers:

- Lemon juice – Adds brightness and acidity.
- Vinegar (any kind) – Adds brightness and acidity.

Managing Texture and Consistency

The texture and consistency of the sauce plays a crucial role in:

- The overall texture of the pizza.

- How the cheese melts.
- The final appearance of the pizza.

Three main factors influence sauce texture and consistency:

1. **The type of tomato product used** – Tomato paste, crushed tomatoes, whole tomatoes, or puree each result in different consistencies.
2. **Thinning or thickening the sauce** – Adjusting the liquid content alters the sauce's consistency.
3. **Processing method** – Blending, straining, or using a food mill affects texture and smoothness.

How Sauce Consistency Affects the Bake and Visuals

Generally, a **thinner** sauce blends more seamlessly with the cheese during baking. As the liquids boil, they help create a uniform mixture of sauce and cheese, often resulting in a smooth, slightly orange-tinted layer (see image below).

Thin sauce, allowing for better integration between the sauce and cheese.

In contrast, a **thicker** sauce does not blend as easily with the cheese, often resulting in a more defined white cheese layer on the pizza:

Thicker sauce – leads to a more distinct separation between the sauce and cheese, often resulting in a noticeable white(er) cheese layer

Sauce consistency also affects **how the cheese bakes**.

A **thinner** sauce **slows down the browning and oiling-off of the cheese**, as the evaporating moisture helps protect it. In contrast, a **thicker** sauce typically causes the cheese to **bake and brown more quickly**.

Simply put, we can use the sauce's consistency as a tool to control how the cheese melts and bakes.

The Effect of the Tomato Product Used on Sauce Texture and Consistency

The type of tomato product determines the sauce's initial texture and consistency out of the can, for example:

- Whole tomatoes crushed by hand – Creates a chunkier sauce, ideal for deep-dish pizzas.
- Tomatoes packed in tomato juice – Can yield a thinner, more watery sauce if all the liquid is used without straining.
- Crushed tomatoes with added tomato paste or puree (passata) – Result in a thicker sauce compared to crushed tomatoes without added paste.

Even among similar products, different manufacturers produce varying consistencies, so adjustments may be needed to reach your desired sauce consistency.

Concentrated tomato products also differ in thickness based on their tomato solids content (measured in degrees Bx), which affects how much thinning is required.

Note that a sauce's texture depends not just on its thickness, but also on its smoothness. The amount of tomato flesh, seeds, and skins present can influence whether a sauce feels chunky or silky smooth; A "thin" sauce isn't necessarily a "watery" sauce.

Achieving the Perfect Thickness (Thinning vs. Thickening)

Adjusting the sauce's consistency is straightforward:

To **thin** the sauce, gradually **add water or tomato juice** until you reach the desired texture.

To **thicken** the sauce, consider the following methods:

- **Straining:** Pass the sauce through a strainer to remove excess liquid. You can then add back some of the liquid to adjust the consistency.
- **Simmering:** Cook the sauce briefly to reduce moisture through evaporation. Keep in mind that this may also concentrate and alter the flavor.

- **Adding a concentrated tomato product:** Mixing in tomato paste or another concentrated product will thicken the sauce. Note that this will also affect the flavor profile.

Tools for Texture: Blenders, Food Mills, and Strainers

For a smoother sauce, you can use a **food mill, strainer, or blender**:

1. Food Mill (Recommended): A food mill is the most effective option, as it removes seeds, skins, unripe tomato pieces, and chunks, resulting in a **completely smooth sauce**. When using a food mill, select a medium-hole disc – neither too large nor too fine.

A food mill also positively affects the sauce's flavor. If you taste the tomato leftovers left in the food mill after processing, you'll notice they taste harsh and bitter.

2. Strainer: A standard strainer can also create a smooth, chunk-free sauce. Simply pour the tomatoes into the strainer and press them through using a silicone spatula, spoon, or even your hand. The finer the strainer mesh, the smoother the sauce.

3. Blender (Least Ideal Option): While a blender is likely the most convenient method, it has several drawbacks:

- Blending does not remove the seeds and skin.
- Air incorporation during blending leads to the oxidation of the tomatoes, resulting in flavor degradation.
- Blending ruptures tomato cells, making the sauce more prone to liquid separation (i.e., more watery).
- The sauce can become airy or frothy, especially on the surface.
- The sauce may shift from deep red to a lighter, pink-ish hue.

If using a blender, **blend in short pulses** and use it as little as possible. When using an immersion blender, keep it **deep in the sauce** rather than near the surface to minimize air incorporation.

Contrary to common belief, tomato seeds, especially when broken during blending, do **not** cause bitterness or significantly affect the sauce's flavor.

Layering Strategy: Sauce Above vs. Below the Cheese

Pizza sauce is typically layered in two primary ways:

- **Traditional Layering (Sauce-First):** Sauce is spread first, followed by cheese and toppings.
- **Upside-Down (Sauce-Over-Cheese):** Cheese is placed first, with the sauce added on top.

Applying the **sauce-over-cheese** method has several effects on the pizza:

- The sauce acts as a protective layer for the cheese, absorbing heat and slowing the cheese's cooking, which affects its **browning and oil release**.
- The pizza develops a **more pronounced red color** on top compared to the sauce-first method (depending on the sauce's color and consistency).
- The sauce undergoes more **direct cooking** due to its increased exposure to heat.
- The sauce's flavor **becomes more prominent**, altering the overall taste balance.
- The cheese acts as a barrier between the dough and the sauce, preventing the dough (to some extent) from absorbing excess moisture. This results in a **drier, crispier crust**.
- The sauce can be applied **sparingly in drizzles or spots**, creating areas with varying sauce intensity for a more dynamic eating (and visual) experience (i.e., different flavors with each bite).

Sauce-over-cheese in a New York style pizza

Sauce-over-cheese in a Sicilian pizza

To summarize: Sauce-over-cheese layering creates a pizza with **a different texture and flavor** compared to the traditional sauce-first layering. It's worth trying to see if you like the result.

Applying sauce over shredded cheese can be challenging, as it doesn't spread evenly across the surface. Because of this, sauce-over-cheese pizzas are often **sauced in a spiral pattern or in scattered dollops** rather than a full, uniform layer.

Alternatively, using **sliced cheese instead of shredded cheese** makes it easier to spread the sauce evenly over the cheese.

To Cook or Not to Cook? (The Pre-Cooking Debate)

Whether or not to cook pizza sauce depends on your desired flavor profile and sauce consistency. However, in most cases and for most applications, **cooking the sauce is unnecessary**.

Traditionally, most pizzas are made with **uncooked** sauce, which is by far the most common type of sauce. This includes some of the most popular pizza styles, such as:
– Neapolitan pizza
– Sicilian pizza
– New York style pizza
– New Haven style pizza
– Chicago deep dish pizza

That said, cooking the sauce isn't off-limits. While less common, some pizzerias use cooked sauce. A notable example is Detroit-style pizza, where the sauce is added **post-bake** and cooked separately, as it doesn't cook on the pizza during baking.

It's important to remember that canned tomatoes are already "cooked" **three times** during the canning process. Then, during baking, they undergo a fourth round of cooking. Additional cooking can alter the sauce's flavor and diminish its natural freshness.

If you're aiming for a deeper, more concentrated flavor, or need to thicken the sauce, cooking can be beneficial. Otherwise, there's no need to cook the sauce. Ultimately, experiment and find what suits your taste.

This only applies to sauces made from "fresh" canned tomatoes, **not** those made with concentrated tomato products. A sauce based on concentrated tomato products doesn't require cooking, as it's already concentrated and intensely flavored, even when diluted with water.

Keep in mind that the amount of sauce displayed is the **final** quantity, after accounting for consistency adjustments (thinning/thickening) and any additional ingredients, not just the weight of the tomato product.

Troubleshooting: Reducing Heartburn and Acidity

Tomatoes are naturally acidic, and acidic foods can contribute to heartburn. If you're prone to heartburn, consider these tips to reduce the risk of heartburn when eating pizza with tomato sauce:

- **Experiment with different tomato products:** Different tomato products have varying acidity levels. If a specific product triggers heartburn, try switching to one made from less acidic tomatoes. You may need to experiment to find the best option for you.
- **Limit dried herbs or switch to fresh herbs:** Dried herbs like oregano, basil, marjoram, and thyme can increase the likelihood of heartburn. If you're sensitive to this, try reducing or omitting dried herbs from your sauce, or consider using **fresh** herbs instead.
- **Incorporate hard cheese into the sauce:** Grate hard cheese such as Parmesan or Pecorino until it's powdery, then mix it into the sauce. The calcium in the cheese helps buffer acidity, potentially reducing the chances of heartburn.

Developing Flavor through Advanced Preparation

The flavor of tomato sauce, regardless of the type, **deepens as it rests for a few hours**, allowing the flavors to meld together.

This is why many sauces, including tomato-based ones, become even more delicious after sitting in the fridge overnight.

As the sauce rests, the flavors have more time to blend, which can noticeably enhance the overall taste. If you compare two identical sauces – one freshly made and one that's rested for a few hours – the difference in flavor is usually evident (though not always dramatic).

For the best results, try making the sauce at least a few hours in advance, ideally the day before or in the morning if you're planning to use it in the evening. Store the sauce in the fridge (or at room temperature If you're using it within a few hours), and take it out before making your pizza.

How Raw Garlic and Onion Affect the Sauce

Tomatoes, like other fruits and plants, contain a soluble fiber called pectin. Fresh garlic and onions contain an enzyme called **pectinase**, which **breaks down** pectin.

When raw garlic or onions are added to a sauce and left **uncooked**, the pectinase enzyme starts to break down the pectin in the tomatoes. Over time, and depending on the quantity of garlic or onion and the temperature at which the sauce is stored, this breakdown will cause the sauce to take on a **jelly-like consistency**.

In simple terms, adding raw garlic or onion to a sauce and letting it sit for an extended period will cause the sauce to become gel-like.

Cooking the sauce with the garlic or onion, or cooking the garlic and onion separately at a temperature above 80°C (176°F), **neutralizes** the pectinase enzyme, preventing the jelly-like transformation.

Enzyme activity is influenced by both concentration and temperature, so the time it takes for the sauce to become jelly-like can vary depending on the amount of garlic or onion and the storage temperature. Generally, this happens within the following time frames:

- 6-12 hours when stored at room temperature
- 12-24 hours when stored in the fridge

For this reason, it's advisable to avoid making a sauce with raw garlic or onion and storing it for extended periods before use. If you're making the sauce in advance for later use, consider the following tips:

- Add raw garlic and/or onion only a couple of hours before using the sauce.
- Store the sauce in the fridge to slow down the break down of pectin.
- Cook the garlic and onions separately before adding them to the sauce (this will change their flavor profile to a sweeter, milder taste – double win).

Storage and Preservation Best Practices

Short-Term Storage

If you have leftover pizza sauce that you plan to use in the short-term – within a week – **store it in an airtight container in the fridge**.

Remember that storing uncooked sauce with onion and/or garlic will cause it to turn into a jelly-like consistency.

Long-Term Storage (Freezing Tomato Sauce)

If you want to store leftover pizza sauce for the long term, **freezing** is the best method. When frozen properly, the sauce will **maintain its original quality** and **stay in good condition for many months**.

Here's how to freeze and store tomato sauce effectively:

1. Pour the sauce into a sealed bag, such as a Ziploc or sandwich bag.
2. Flatten the bag, removing as much air as possible, and seal it tightly.
3. Transfer the flat bag to the freezer.

While you can use other containers to store the sauce in the freezer, using a bag has several advantages:

- **Space Efficiency:** The flat shape minimizes space usage, allowing you to stack multiple bags on top of each other
- **Portion Control:** This method lets you prepare individual portions – each bag can contain one pizza's worth of sauce, so you only thaw what you need.
- **Faster Freezing:** The larger surface area (due to the flattening of the sauce) allows the sauce to freeze faster, minimizing damage to the tomato cells, and preserving the sauce's quality.
- **Reduced Air Exposure:** Flattening the bag and removing air helps maintain the sauce's texture and flavor, unlike bulkier containers that typically leave a layer of air inside.

To use the sauce, simply transfer it from the freezer to the fridge to **thaw overnight**. While you *can* thaw it at room temperature directly from the freezer, the quicker thawing can make the sauce watery and affect its quality. For best results, thaw the sauce in the fridge the night before you plan to use it.

After thawing, the sauce may be thinner and more watery than before freezing, as the freezing process can cause tomato cells to break down and release water.

To 'fix' the consistency of the sauce, strain some of the excess liquid using a strainer, or pour the sauce to a bowl and scoop out excess water with a spoon or teaspoon – most of the water will accumulate at the top.

Pizza Tomato Sauce: Final Summary

Tomato sauce for pizza is an art form in itself. Every component of the sauce – from the type of tomato product, to the preparation method, seasonings, consistency, and amount used – directly influences the taste, texture, and overall quality of the pizza.

To find the sauce that best suits *your* preferences, there's no substitute for **experimentation**. You'll need to try different products, quantities, seasonings, and consistencies.

Here are some suggestions for creating a base sauce, which can serve as a great starting point:

- **Crushed tomatoes or tomato puree** straight from the can form an excellent foundation and offer a good benchmark for comparison with other sauces.
- **Whole tomatoes in juice** – straining the tomatoes (keeping the tomato juice), processing them into a sauce consistency using a blender, food mill, or strainer, and then adding back the tomato juice to reach the desired thickness.
- **Tomato paste** can be turned into a sauce by adding water until it reaches the consistency you like.
- **A mix of crushed or whole tomatoes with tomato paste** can also work well, with a paste-to-tomato ratio ranging from 5% to 50% (e.g., 5% paste and 95% crushed tomatoes).

Or, you can try PizzaBlab's pizza sauce recipe (recommended).

Start with minimal seasoning and taste as you go, adjusting the flavors to your liking. From there, experiment with different combinations of tomato products, textures, consistencies and the amount of sauce you apply to your pizza – the possibilities are endless.

Key Takeaways

- **Beyond Just a Recipe:**
  This guide goes beyond a simple recipe to explore the underlying technical aspects and techniques of tomato sauce. It provides the fundamentals you need to build a sauce from the ground up, tailored to your specific flavor profile and texture goals.
- **The Umami Powerhouse:**
  Tomatoes are naturally high in glutamic acid (umami), which is why the sauce often dominates the overall flavor profile of the pizza.

- **The Four Main Sauce Profiles:**
  Pizza sauces generally fall into four categories based on whether they are **plain vs. seasoned** or **fresh (uncooked) vs. concentrated (cooked)**. Choosing between these allows you to match the sauce to your desired flavor profile.
- **The Canned Gold Standard:**
  Canned tomatoes are the professional choice for pizza sauce. Because they are vine-ripened and processed within hours of harvest, they offer a consistent quality and flavor that "fresh" supermarket tomatoes (which are often picked green) cannot match.
- **Seasoning Balance:**
  A "plain" sauce typically relies on high-quality canned tomatoes and salt alone, while a "seasoned" sauce uses herbs and spices to build complexity, or balance out less flavorful tomatoes.
- **Tomato Product Variety:**
  From whole peeled tomatoes and crushed tomatoes to tomato purée and paste, each product offers a different starting point for flavor intensity and texture.
- **Impact on the Bake:**
  The moisture content and consistency of your sauce are critical to the final result. A sauce that is too watery or thin can release excess moisture into the dough, leading to a "gum line" (a layer of raw, undercooked dough directly under the sauce). Additionally, the thickness of the sauce dictates how the cheese melts and browns, directly influencing the final visual appeal of the pizza.
- **To Cook or Not to Cook:**
  This decision depends entirely on your target flavor profile. Cooking the sauce beforehand creates a deep, rich, and highly concentrated taste, but at the expense of the tomato's natural brightness and "freshness." If you prefer a vibrant, zesty sauce, it is best to avoid pre-cooking. Always keep in mind that the sauce will undergo further cooking and reduction once it enters the high-heat environment of the oven.

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